- Homogenius, isotropic, unbound, source free ($S=J=0$)

a) $\bar{\nabla}\cdot\bar{B}=0$ b) $\bar{\nabla}\cdot\bar{D}=0$ c) $\bar{\nabla}\times\bar{E}=-\frac{d\bar{B}}{dt}$ d) $\bar{\nabla}\times\bar{H}=\frac{d\bar{D}}{dt}$

e) $\bar{B}=\mu\bar{H}$ f) $D=\varepsilon\bar{E}$

- $\nabla^2 E=\mu\epsilon\frac{d^2E}{dt^2}=-\omega^2\mu\epsilon\bar{E}$

$\nabla^2 H=\mu\epsilon\frac{d^2H}{dt^2}=-\omega^2\mu\epsilon\bar{H}$

- uniform $\bar{E}$ in space $\Rightarrow \bar{E}=\bar{K_0} \Rightarrow \bar{E}\equiv 0$
- $\bar{E}=E_0\hat{x} \Rightarrow \bar{H}\equiv 0 \Rightarrow \bar{E}=0$
- $\bar{E}=E_0(z)\hat{x} \Rightarrow \frac{dE_0(z)}{dz}\hat{y}=-j\omega\mu\bar{H} \Rightarrow \bar{H}=\frac{-1}{j\omega\mu}\frac{dE_0(z)}{dz}$

$$\frac{d^2E(z)}{dz^2}=-\omega^2\mu\epsilon E(z)$$

(propogation constant)$^2=\gamma^2$

$\gamma=j\omega\sqrt{\mu\epsilon}$; $\beta=\omega\sqrt{\mu\epsilon}$ (phase constant)

$\alpha=0$ (attenuation constant)

$\Rightarrow$ lossless

$$\Rightarrow \bar{E}(z)=\left[E_x^+ e^{j\beta z}+E_x^- e^{j\beta z}\right]\hat{x}$$

$$\Rightarrow \bar{H}(z)=\left[\frac{\beta}{\mu\omega}E_x^+ e^{-j\beta z}-\frac{\beta}{\mu\omega}E_x^- e^{j\beta z}\right]\hat{y}$$

- intrinsic impedance $\eta=\frac{\omega\mu}{\beta}=\sqrt{\frac{\mu}{\epsilon}}$ η_0 (free space) $=120\pi\ \Omega$

E, H, wave direction $\longrightarrow$ right handed triad

$\frac{|E|}{|H|}=\eta$ $\bar{H}=\frac{1}{\omega\mu}(\bar{k}\times\bar{E})$

- Polarization $\rightarrow$ direction to electric field

treat $\bar{E}$ like an arrow, shape of which the tip of the $\bar{E}$ vector draws in a plane $\perp$ to direction of wave propogation as function of time

- $E_x = E_1 \cos\omega t \quad E_y = E_2 \cos(\omega t + \delta)$

  locus of tip of $\bar{E}$ field

$$\frac{E_x^2}{E_1^2} - \frac{2E_xE_y}{E_1E_2}\cos\delta + \frac{E_y^2}{E_2^2} = \sin^2\delta$$

$\delta = 0 \Rightarrow E_y = E_x \frac{E_2}{E_1}$ Straight line → linear polarization

$\delta = \pm\pi/2$, $E_1 = E_2 = E_0 \Rightarrow E_x^2 + E_y^2 = E_0^2$ → circular polarization

left circular polarization: $\delta = +\pi/2$

right circular polarization: $\delta = -\pi/2$

$\delta \neq \pi/2$; $E_1 \neq E_2 \Rightarrow$ Elliptical polarization

τ → tilt angle } wave parameters

AR → a/b (axial ratio)

- State of polarization

  Electrical parameters: E_2/E_1, δ; $\gamma = \tan^{-1}(E_2/E_1)$

  wave parameters: $\pm AR$, τ; $\epsilon = \cot^{-1}(\pm AR)$

$\gamma \in [0, \pi]$ $\delta \in [-\pi/2, \pi/2]$ $\epsilon \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ $\tau \in [0, \pi]$

- $\nabla \times \bar{H} = j\omega\epsilon_0\{\epsilon_r - j\frac{\sigma}{\omega\epsilon_0}\}\bar{E}$

  relative permittivity (complex dielectric const) ϵ_{rc}

- If conduction current >> Displacement current ($\sigma >> \omega\epsilon$) → conductor

  If conduction current << Displacement current ($\sigma << \omega\epsilon$) → dielectric

$\sigma = \omega\epsilon = \omega\epsilon_0\epsilon_r \Rightarrow \omega_T = \frac{\sigma}{\epsilon_0\epsilon_r}$

$\omega > \omega_T \rightarrow$ dielectric $\quad \omega < \omega_T \rightarrow$ conductor

- $\nabla^2 \begin{Bmatrix} \bar{E} \\ \bar{H} \end{Bmatrix} = -\omega^2 \mu\epsilon_0\epsilon_{rc} \begin{Bmatrix} \bar{E} \\ \bar{H} \end{Bmatrix} \rightarrow \gamma = j\omega\sqrt{\mu\epsilon_0\epsilon_{rc}}$

$$\alpha = Re(\gamma) = \omega\sqrt{\frac{\mu_0\epsilon_0\epsilon_r}{2}}\left\{\sqrt{1+\frac{\sigma^2}{\omega^2\epsilon_0^2\epsilon_r^2}} - 1\right\}^{1/2}$$

$$\beta = Im(\gamma) = \omega\sqrt{\frac{\mu_0\epsilon_0\epsilon_r}{2}}\left\{\sqrt{1+\frac{\sigma^2}{\omega^2\epsilon_0^2\epsilon_r^2}} + 1\right\}^{1/2}$$

- loss tanget: $\tan\delta = \frac{\sigma}{\omega\epsilon} << 1$ (for good dielectric
- Good dielectric / low-loss medium: $\omega\epsilon >> \sigma$

$$\Rightarrow \alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} \qquad \beta \approx \omega\sqrt{\mu\epsilon}$$

- intrinsic impedance $\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$

- Good conductor: $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \approx \sqrt{j\omega\mu\sigma}$

$$\alpha = \sqrt{\frac{\mu\omega\sigma}{2}} = \beta$$

- wave amplitude $\sim e^{-\alpha x}$

skin depth $\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}} = \frac{1}{\sqrt{\pi f\mu\sigma}}$ (penentration)

- Good conductor: $\eta = \sqrt{\frac{\omega\mu}{\sigma}}\angle 45^\circ = \frac{|E|}{|H|}$ ($\bar{H}$ lags by 45°)

- $\bar{E} = \bar{E_0} e^{j(\omega t - \beta x)}$ velocity of wave $V = \frac{\omega}{\beta}$

$V(\text{medium}) = \frac{c}{\sqrt{\epsilon_r}}$ Refractive index $n = \sqrt{\epsilon_r}$

conductor: $v_p = \frac{\omega}{\sqrt{\omega\mu\sigma/2}} = \sqrt{\frac{2\omega}{\mu\sigma}}$ $v_p \sim$ frequency

↓ dispersive medium

- $\bar{\nabla} \times \bar{E} = -\mu\frac{d\bar{H}}{dt} \qquad \bar{\nabla} \times \bar{H} = \bar{J} + \epsilon\frac{d\bar{E}}{dt}$

$$\nabla \cdot (\bar{E} \times \bar{H}) = -\frac{\mu}{2}\frac{\partial}{\partial t}|\bar{H}|^2 - \frac{\epsilon}{2}\frac{\partial}{\partial t}|\bar{E}|^2 - \bar{E}\cdot\bar{J}$$

integrate over a vol.

$$\Rightarrow \int_S (\bar{E}\times\bar{H})\cdot d\bar{a} = -\frac{\partial}{\partial t}\int_V \frac{\mu}{2}|\bar{H}|^2 dV - \frac{\partial}{\partial t}\int_V \frac{\epsilon}{2}|\bar{E}|^2 dV - \int_V \bar{E}\cdot\bar{J}\,dV$$

↓ Net power flow from a closed surface

Poynting theorm

- $\bar{P} = \bar{E}\times\bar{H}$ ($P \rightarrow$ poynting vector)
- $\bar{E} = \bar{E}_0 e^{j(\omega t+\phi_e)}$ $\quad \bar{H} = \bar{H}_0 e^{j(\omega t+\phi_h)}$

Instantaneous values of the fields:

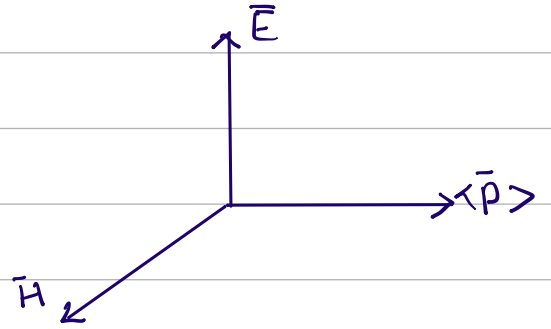
$$\bar{E}(t) = Re\{E_0 e^{j\omega t} e^{j\phi_e}\}\cdot\hat{e} = E_0\cos(\omega t+\phi_e)\hat{e}$$

$$\bar{H}(t) = Re\{H_0 e^{j\omega t} e^{j\phi_h}\}\cdot\hat{h} = H_0\cos(\omega t+\phi_h)\hat{h}$$

$$\bar{P} = \bar{E}\times\bar{H} = \frac{E_0 H_0}{2}\{\cos(\phi_e-\phi_h)+\cos(2\omega t+\phi_e+\phi_h)\}(\hat{e}\times\hat{h})$$

$$\langle\bar{P}\rangle = \frac{1}{T}\int_0^T \bar{P}\,dt = \frac{1}{2}Re\{\bar{E}\times\bar{H}^*\} \quad (always >0)$$

- Uniform plane wave:



$\frac{E}{H} = \eta$ (intrinsic impedance)

$$\langle\bar{P}\rangle = \frac{1}{2}Re\left\{|E_0|^2\frac{1}{\eta^*}\right\} = \frac{1}{2}Re\{\eta|H_0|^2\}$$

dielectric medium: $\eta = \sqrt{\mu/\epsilon}$ (real) $\Rightarrow \langle\bar{P}\rangle = \frac{1}{2}|E_0|^2/\eta$

good conductor: $\eta \approx \sqrt{\frac{\omega\mu}{2\sigma}} + j\sqrt{\frac{\omega\mu}{j\sigma}} \Rightarrow \langle\bar{P}\rangle = \frac{1}{2}|E_0|^2\sqrt{\frac{\sigma}{2\omega\mu}}$

- Conduction current density $\bar{J}(z) = \sigma\bar{E} = \sigma E_0 e^{-\gamma z}\hat{x}$

$$\bar{I}(z) = \bar{J}(z)\,dz = \sigma E_0 e^{-\gamma z}dz\,\hat{x}$$

$$\Rightarrow I(z)\ (\text{surface current}) = \int_0^\infty \bar{I}(z) = \frac{\sigma E_0}{\gamma}\hat{x}$$

$\eta_c = \sqrt{\frac{j\omega\mu}{\sigma}} = \gamma/\sigma$; $H_0 = E_0/\eta_c \Rightarrow |H_0| = |\bar{I}(z)|$

- Surface impedance: $Z_s = \frac{\bar{E}_{tangential}}{J_s} = \eta_c$ (for good conductor)

Ohmic power loss in $1m^2$ slab, thickness dz

$$dw = \frac{1}{2}|I|^2 R = \frac{1}{2}\sigma E_0^2 e^{-2\alpha z}\, dz$$

Total power loss (unit area) $w = \frac{\sigma}{4\alpha}|E_0|^2 = \frac{\sigma}{\sqrt{2}\gamma}|E_0|^2$

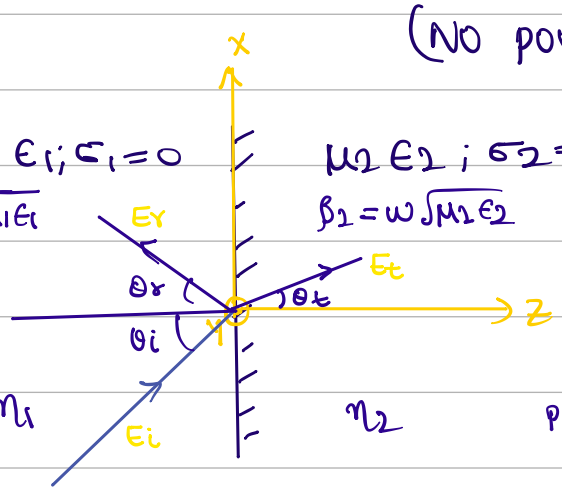
$$= \frac{1}{2}|J_s|^2\sqrt{\frac{\omega\mu}{2\sigma}}$$

Poynting vector $\bar{P} = E_0 H_0^* \hat{z} = \frac{|E_0|^2}{2\eta_c^*} = \frac{1}{2}|E_0|^2\sqrt{\frac{2}{\omega\mu\sigma}}$

(No power flow at $z=\infty$) $P \propto 1/\sqrt{\sigma}$, $P \propto 1/\sqrt{f}$

- $\mu_1 \epsilon_1; \sigma_1 = 0$ $\quad \mu_2 \epsilon_2; \sigma_2 = 0$

$\beta_1 = \omega\sqrt{\mu_1\epsilon_1}$ $\quad \beta_2 = \omega\sqrt{\mu_2\epsilon_2}$



$$\bar{E}_i = E_{0i} e^{-j\beta_1(\cos\phi_x x + \cos\phi_y y + \cos\phi_z z)}$$
$$= F_{0i} e^{-j\beta_1(x\sin\theta_i + z\sin\theta_i)}$$

magnitude of field in xy plane ($z=0$)

$$= Re\{F_i\} = F_{0i}\cos(\beta_1 x \sin\theta_i)$$

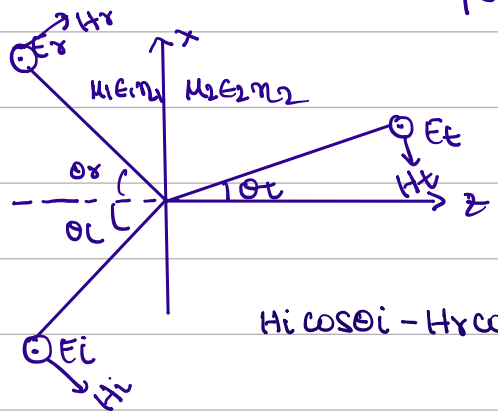
phase gradient: $-\beta_1 \sin\theta_i$

$\theta_r = \theta_i$; $\sqrt{\mu_1\epsilon_1}\sin\theta_i = \sqrt{\mu_2\epsilon_2}\sin\theta_t$; $\sqrt{\epsilon_r} = n_r$

$\Gamma = \frac{E_r}{E_i}$ $\quad \tau = \frac{E_t}{E_i}$

parallel polarization: E in plane of incidence

perpendicular polarization: $E \perp$ to plane of incidence



$$E_i = E_{i0} e^{j\beta_1(x\sin\theta_i + z\cos\theta_i)}\hat{y}$$
$$E_r = E_{r0} e^{-j\beta_1(x\sin\theta_i - z\cos\theta_i)}\hat{y}$$
$$E_t = E_{t0} e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}\hat{y}$$

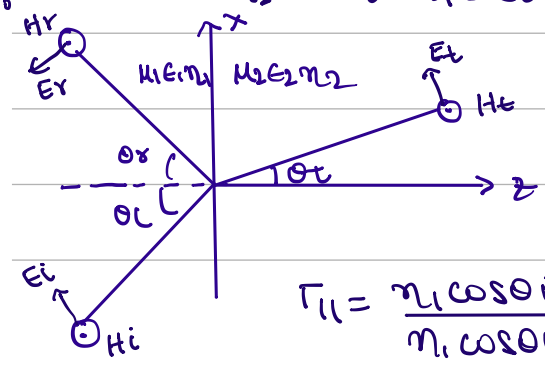
at $z=0$, $E_{i0} e^{-j\beta_1 x\sin\theta_i} + E_r e^{-j\beta_1 x\sin\theta_i} = E_t e^{-j\beta_2 x\sin\theta_t}$

$$\Rightarrow \boxed{E_i + E_r = E_t}$$

$H_i\cos\theta_i - H_r\cos\theta_i = H_t\cos\theta_t$

$$\Rightarrow \boxed{\frac{E_i}{\eta_1}\cos\theta_i - \frac{E_r}{\eta_1}\cos\theta_i = \frac{E_t}{\eta_2}\cos\theta_t}$$

$$\Gamma_\perp = \frac{\eta_2\sec\theta_t - \eta_1\sec\theta_i}{\eta_2\sec\theta_t + \eta_1\sec\theta_i} \; ; \; \tau_\perp = \frac{2\eta_2\sec\theta_t}{\eta_2\sec\theta_t + \eta_1\sec\theta_i} \; ; \; 1 + \Gamma_\perp = \tau_\perp$$



$H_i + H_r = H_t \Rightarrow \boxed{\frac{E_i}{\eta_1} + \frac{E_r}{\eta_1} = \frac{E_t}{\eta_2}}$

$$\boxed{E_i\cos\theta_i - E_r\cos\theta_i = E_t\cos\theta_t}$$

$$\Gamma_{11} = \frac{\eta_1\cos\theta_i - \eta_2\cos\theta_t}{\eta_1\cos\theta_i - \eta_2\cos\theta_t} \; ; \; \tau_{11} = \frac{2\eta_2\cos\theta_i}{\eta_1\cos\theta_i - \eta_2\cos\theta_t} \; ; \; 1 + \Gamma_{11} = \frac{\eta_1}{\eta_2}\tau_{11}$$

- Total internal reflection: $\beta_1/\beta_2 \sin\theta_i > 1 \Rightarrow \theta_c$: θ_i such that $\theta_t = \pi/2$

  $\sin\theta_c = \sqrt{\mu_2\epsilon_2}/\sqrt{\mu_1\epsilon_1}$

  (there's a phase change while TIR reflections)

$$\phi_{\parallel} = -2\tan^{-1}\left(\frac{n_2\sqrt{\beta_1^2/\beta_2^2 \sin^2\theta_i - 1}}{n_1\cos\theta_i}\right) \qquad \phi_{\perp} = -2\tan^{-1}\left(\frac{n_1\sqrt{\beta_1^2/\beta_2^2 \sin^2\theta_i - 1}}{n_2\cos\theta_i}\right)$$

(Evanescent fields)

Fields in medium 2: $E_t = E_{to}\, e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$

(no power flow, but ($\tau \neq 0$) fields exist) $= E_{to}\, \underbrace{e^{-z(\beta_1^2\sin^2\theta_i - \beta_2^2)^{1/2}}}_{\text{decaying field}}\, \underbrace{e^{-jx\beta_1\sin\theta_i}}_{\text{travelling wave in x-direction}}$

- linearly polarized $\bar{E}$ ($\phi = 0°$):
  - → normal reflection: $\Gamma_{\parallel}, \Gamma_{\perp} \rightarrow$ real, $|\Gamma_{\perp}| \neq |\Gamma_{\parallel}| \rightarrow$ linearly polarized
  - → TIR: $\Gamma_{\parallel}, \Gamma_{\perp} \rightarrow$ img, $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1 \rightarrow$ eliptical polarized
- Circular polarized $\bar{E}$ ($\phi = 90°$)
  - → normal reflection: $\Gamma_{\parallel}, \Gamma_{\perp} \rightarrow$ real, $|\Gamma_{\perp}| \neq |\Gamma_{\parallel}| \rightarrow$ eliptical (tilt angle = 0 or $\pi/2$)
  - → TIR: → elliptical polarized
- $\Gamma_{\parallel} = 0 \Rightarrow n_1\cos\theta_i = n_2\cos\theta_t \Rightarrow \tan\theta_B = \beta_2/n_2\left(\frac{n_1^2 - n_2^2}{\beta_2^2 - \beta_1^2}\right)^{1/2}$

  $\Gamma_{\perp} = 0 \Rightarrow n_1\cos\theta_t = n_2\cos\theta_i \Rightarrow \tan\theta_B = \frac{\beta_2}{n_1}\left(\frac{n_2^2 - n_1^2}{\beta_2^2 - \beta_1^2}\right)^{1/2}$

  $\theta_B \rightarrow$ Brewster angle

  $\theta_i = \theta_{B(\parallel)} \Rightarrow E_{r\perp}$ will be linearly polarized $\perp$ to POI